

Module 3

Business Economics

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer 1(a)(i)

In equilibrium,

- $150 + 30P^* = 500 - 40P^*$
- $P^* = \text{HK\$}5$
- $Q^* = 300$ (using either demand or supply function)

Answer 1(a)(ii)

At HK\$4, $Q(s) = 150 + 30(4) = 270$

$Q(d) = 500 - 40(4) = 340$

Create Shortage (Excess demand) of 70 units

Answer 1(a)(iii)

The two forms of price control are:

- price ceiling (imposed maximum price)
- price floor (imposed minimum price)

Answer 1(a)(iv)

If the market structure of wheat allows the supplier(s) with strong price-setting power to exploit consumers, then the government intervention may be one of the ways to deal with the inefficient allocation of scarce resources problem.

A black market may develop for people who do not get the commodity at the controlled price (the “ceiling price”) but who are willing to pay a (black) market price that is higher than the controlled price.

Answer 1(b)(i)

Gasoline and car are complementary goods.

The increase in price of gasoline will decrease the demand for cars.

The demand curve of automobiles will shift to the left.

Both the equilibrium price and quantity of automobiles will fall.

Answer 1(b)(ii)

Steel is a major input for production of cars.

If the cost of steel increases, then this will increase the cost of producing cars.

The supply curve of automobiles will shift to the left.

Equilibrium price will rise and the equilibrium quantity will fall.

Answer 2(a)

Gains from international trade refer to the net benefits to countries from being allowed to engage in voluntary trading with each other.

Sources of gains from (international) trade include:

- Specialisation/ division of labour
- Expanded market size
- Economies of scale
- Economies of scope

[Any two of the above]

[Other valid points are acceptable]

Answer 2(b)

- A tariff is a tax on imports imposed by the importing country. A quota, on the other hand, is a maximum limit on the quantity of goods that can be imported into a country.
- Tariffs make imports more expensive and hence less competitive. In the case of quota, less import will raise the equilibrium price of imports.

Answer 2(c)(i)

A standard measure of an economy's degree of trade openness is the ratio of the sum of exports and imports to its GDP. The higher the value is, the more economically open the economy is.

Answer 2(c)(ii)

Given the size and diversity of the structure of Hong Kong's economy, it has a much higher value than the US based on the measure.

Answer 2(c)(iii)

Fixed exchange and floating exchange rate systems (regimes).

Answer 2(c)(iv)

Hong Kong's linked exchange rate system is one type of fixed exchange rate system.

Answer 2(c)(v)

Advantages:

- Enhance international investors' confidence towards the Hong Kong dollar which is crucial for Hong Kong as an international financial centre.
- Keep Hong Kong's banking system and government economic policy more stable.

Disadvantages:

- The loss of independent monetary policy means that Hong Kong has less means to defend the economy from external shocks when they occur.
- With the economic cycles of Hong Kong and the US not necessarily being consistent, the US monetary policy (which Hong Kong has to follow) may at times aggravate the territory's economic downturn.

[Other valid arguments are acceptable]

Answer 3(a)

$$\text{GDP Deflator (2019)} = \left[\frac{\$9 \times 100 + \$10 \times 200}{\$9 \times 100 + \$10 \times 200} \right] \times 100 = 100$$

$$\text{GDP Deflator (2020)} = \left[\frac{\$10 \times 130 + \$12 \times 220}{\$9 \times 130 + \$10 \times 220} \right] \times 100 = 116.91$$

$$\text{Consumer Price Index (2019)} = \left[\frac{\$9 \times 100 + \$10 \times 200}{\$9 \times 100 + \$10 \times 200} \right] \times 100 = 100$$

$$\text{Consumer Price Index (2020)} = \left[\frac{\$10 \times 100 + \$12 \times 200}{\$9 \times 100 + \$10 \times 200} \right] \times 100 = 117.24$$

Answer 3(b)

$$\% \text{ Change in P (GDP Deflator)} = \frac{116.91}{100} - 1 = 0.1691 \text{ (16.91\%)}$$

$$\% \text{ Change in P (CPI)} = \frac{117.24}{100} - 1 = 0.1724 \text{ (17.24\%)}$$

CPI-based price change is larger than that of GDP Deflator because the weighting of Bradbury hazelnut chocolate bars in the basket in the base year (which is fixed over time) is larger than that of Venus chocolate bars and the (percentage in the) price increase of Bradbury hazelnut chocolate bars is also larger than that of Venus chocolate bars.

Answer 3(c)

Neither of the price indexes is considered clearly superior by economists.

Consumer Price Index (based on base year quantity) tends to overstate the increase in the cost of consumption by failing to capture consumers' substitution of cheaper products for expensive ones.

On the other hand, the GDP Deflator (based on current year quantity) tends to understate the increase in the cost of consumption, as it fails to take into account the fact that the induced substitutions would make consumers worse off.

Answer 4(a)(i)

Inferential statistics refers to the practice of using a sample of data to draw conclusions about the unknown characteristics of the population.

Random sampling is a procedure for sampling from a population in which the members of the sample are selected by chance, with every member has a known, non-zero probability of being selected.

Answer 4(a)(ii)

The advantages of this method are that it:

- ensures the sample is representative of the target population; and
- is the simplest form to collect data.

The disadvantages are that it is:

- requires the researcher to be experienced in data collection (so as to separate accurate data collected from inaccurate data); and
- incurs higher monetary and time costs (as every subject in the population must be interviewed or reviewed).

[Other valid answers are acceptable]

Answer 4(a)(iii)

Probability sampling methods:

- Simple random sampling: Each unit in the population has an equal chance of being selected.
- Systematic sampling: Each member of the entire population is first numbered and then sample members are chosen based on a random starting point but with a fixed interval.
- Cluster sampling: Divides a population into groups (clusters) that are representative of the population and then studies the selected clusters.
- Stratified sampling: Divides a population into groups (strata) that share a characteristic of interest and then draws a random sample from each strata.

[Any two of the above]

[Other valid methods are also acceptable]

Answer 4(b)

A public good has the defining characteristics of being non-excludable and non-rivalrous.

Non-excludable means it is impossible or prohibitively costly for the owner to exclude others from using the good. Non-rival means that one person's use of a good does not preclude others from using it.

Answer 4(c)

Jocelyn's first choice is to work full time as a consultant. Her next best opportunity/alternative is teaching summer courses and travelling for the remainder of the summer.

Therefore, Jocelyn's opportunity cost of being a consultant is the HK\$20,000 she would have earned from teaching, as well as the satisfaction she could have derived from travelling.

* * * END OF EXAMINATION PAPER * * *